CS1632: Nondeterminism and QA

Wonsun Ahn

University of Pittsburgh

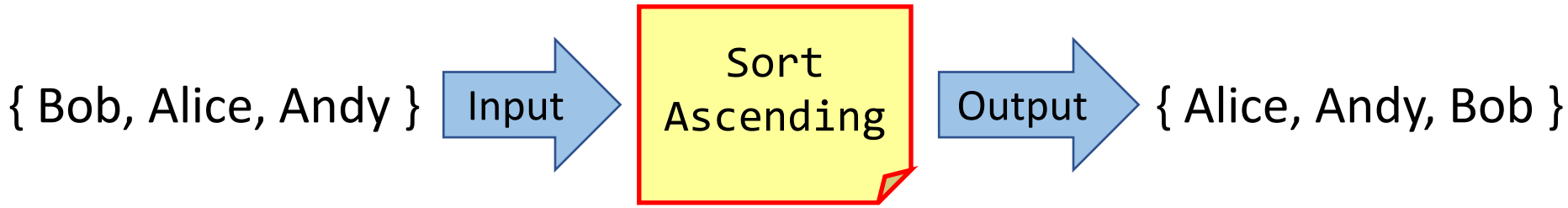
Learning Goals

- Learn what **nondeterminism** is and why it is an issue with SW testing
- Learn **4 sources** of nondeterminism
- Learn **3 tools** that combat nondeterminism

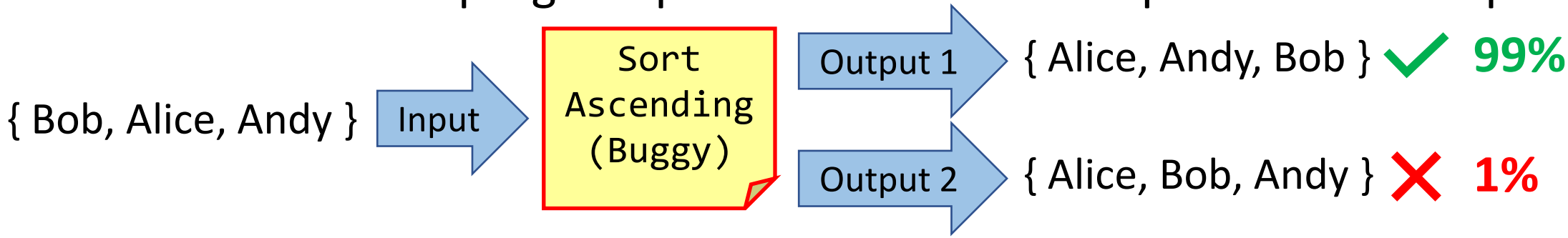
- To understand this fully, it helps to know:
  - C programming
  - Concept of program stack and heap
  - Concept of threads and parallel execution
  - If you don’t, I will go slowly on some concepts so don’t worry

What is Nondeterminism?

- When the output of a program is not determined by its input
- A deterministic program produces the same output on the same input:

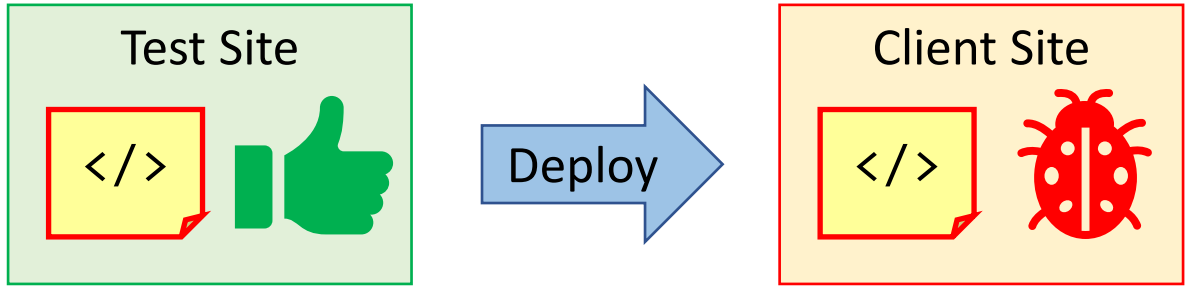


- A nondeterministic program produces different outputs on same input!



Nondeterminism Makes Testing Hard

- Surprise defects
  - Defect not revealed during testing suddenly pops up during usage



- Unreproducible defects
  - Defect revealed during testing doesn't show up when trying to debug it



University of Pittsburgh

So What To Do?

- Depends on what kind of nondeterminism it is.

1. Nondeterminism by mistake
   - Coder never intended nondeterminism; nondeterminism itself is the defect

   → Stamp out the nondeterminism!

2. Nondeterminism by design
   - Coder intended the nondeterminism

   → Must somehow deal with the nondeterminism.

Outline

- Nondeterminism by mistake
  - Memory errors (examples / solutions)
  - Datarace errors (examples / solutions)
- Nondeterminism by design
  - Thread interleaving (examples / solutions)
  - Random number generation (examples / solutions)
- Summary

Nondeterminism by Mistake

It’s a Mistake – Stamp out from your Code!

- Due to erroneous code
  - Memory errors
  - Datarace errors
- Runtime behavior of program is undefined or barely defined
  - Called errors because they are illegal in language specification
- Undefined behavior can hardly be intentional by design
  - These behaviors need to be banished!

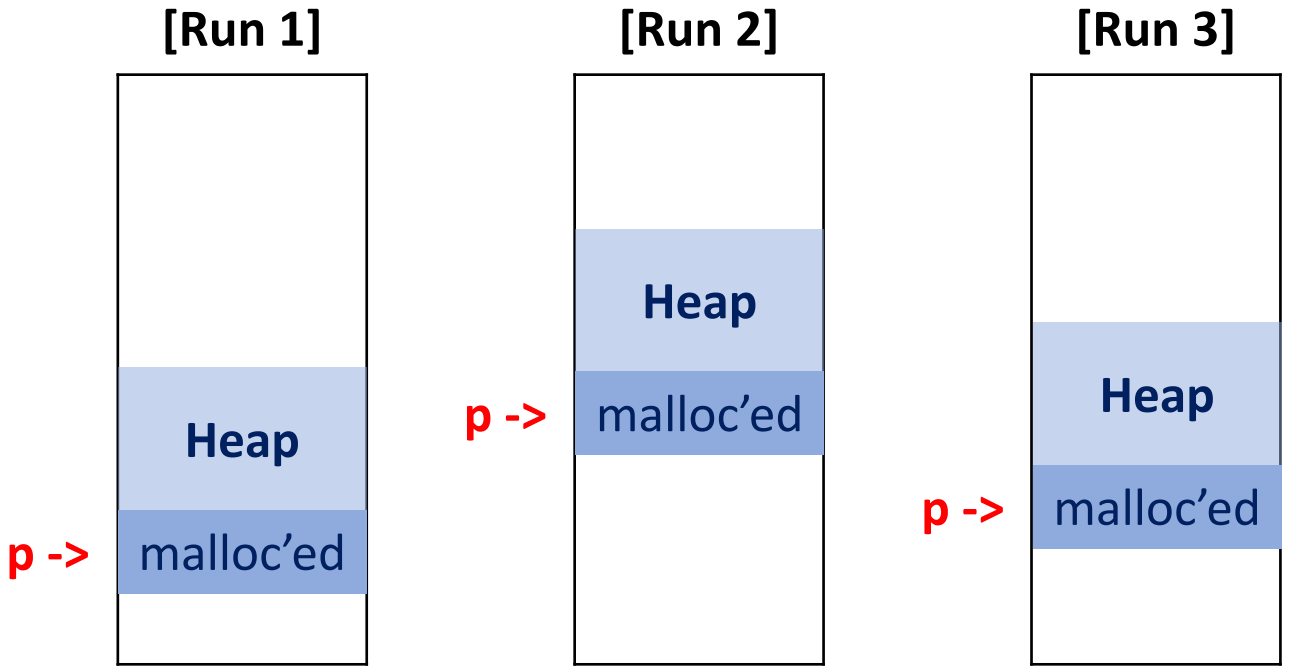
Outline

- Nondeterminism by mistake
  - Memory errors (examples / solutions)
  - Datarace errors (examples / solutions)
- Nondeterminism by design
  - Thread interleaving (examples / solutions)
  - Random number generation (examples / solutions)
- Summary

It’s Very Easy to Make a Random C Program

```
int main() {
   char *p = malloc(8);
   printf("p = %p\n", p);
   free(p);
   return 0;
}
```

```
bash-4.2$ ./heap
p = 0x10b2010
bash-4.2$ ./heap
p = 0x257e010
bash-4.2$ ./heap
p = 0x13a7010
```

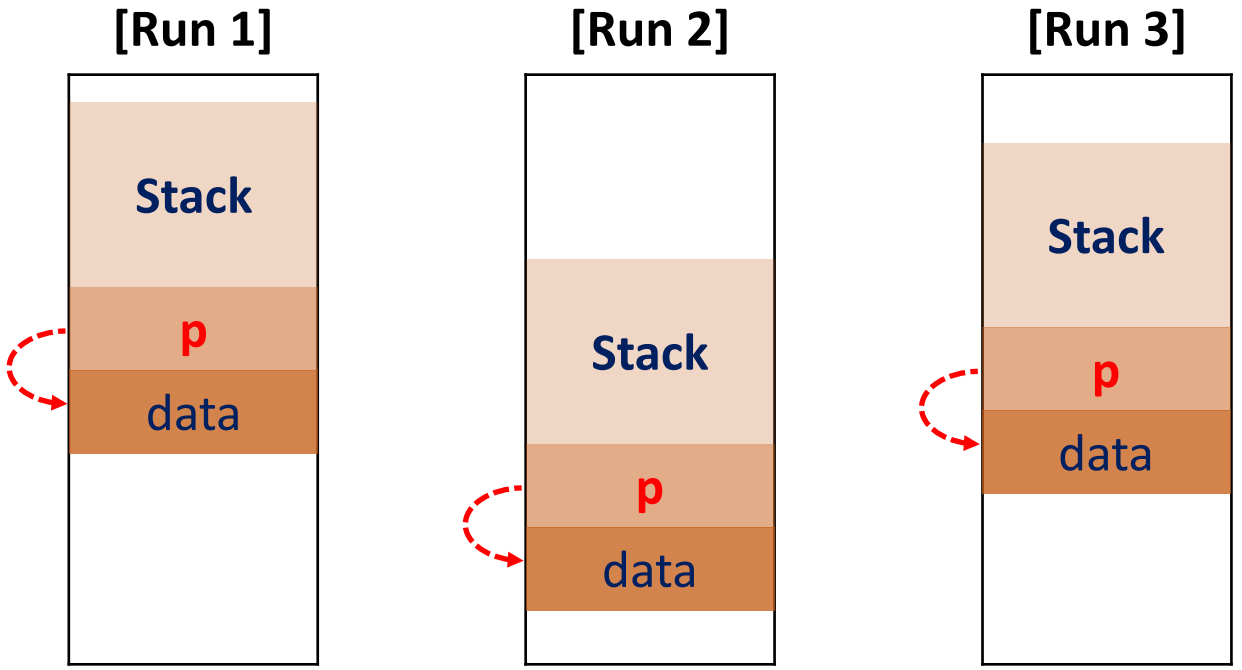


- Why is malloc returning a random address?
- Address Space Layout Randomization (ASLR)
- To prevent hackers from guessing memory layout

Stack Addresses are Random Too

```
int main() {
   char *p;
   char data[8];
   p = data;
   printf("p = %p\n", p);
   return 0;
}
```

```
bash-4.2$ ./stack
p = 0x7ffff5443188
bash-4.2$ ./stack
p = 0x7ffedfb740f8
bash-4.2$ ./stack
p = 0x7fffc21002f8
```



- `p` is now pointing to a stack location
- Why is `data[8]` at a random address?
- ASLR is also applied to the stack

Does it Matter?

- Aren't these contrived examples?
- Pointer addresses are almost never part of program output anyway
  - Unless you are printing them out for debugging or diagnostic purposes
  - Most of the times you output the data stored inside those locations (e.g. You output the data inside a data structure node not the node address)
- But addresses can leak out to program output *by mistake*
- Specifically when you have *memory errors*

Memory Errors

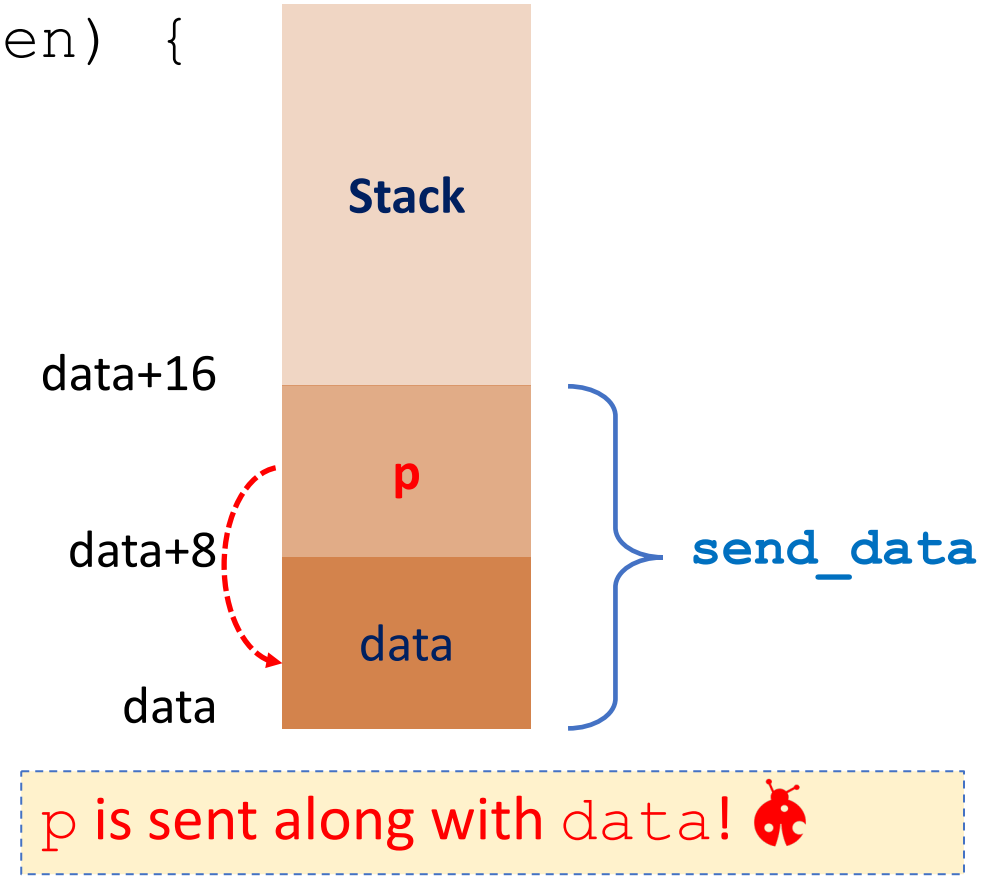
- Errors that access an illegal memory location are the culprits
  - Buffer overflow: access beyond the bounds of an array
  - Dangling pointer: access to already freed memory pointed to by pointer

- If illegal location contains an address, it can leak out to the output!

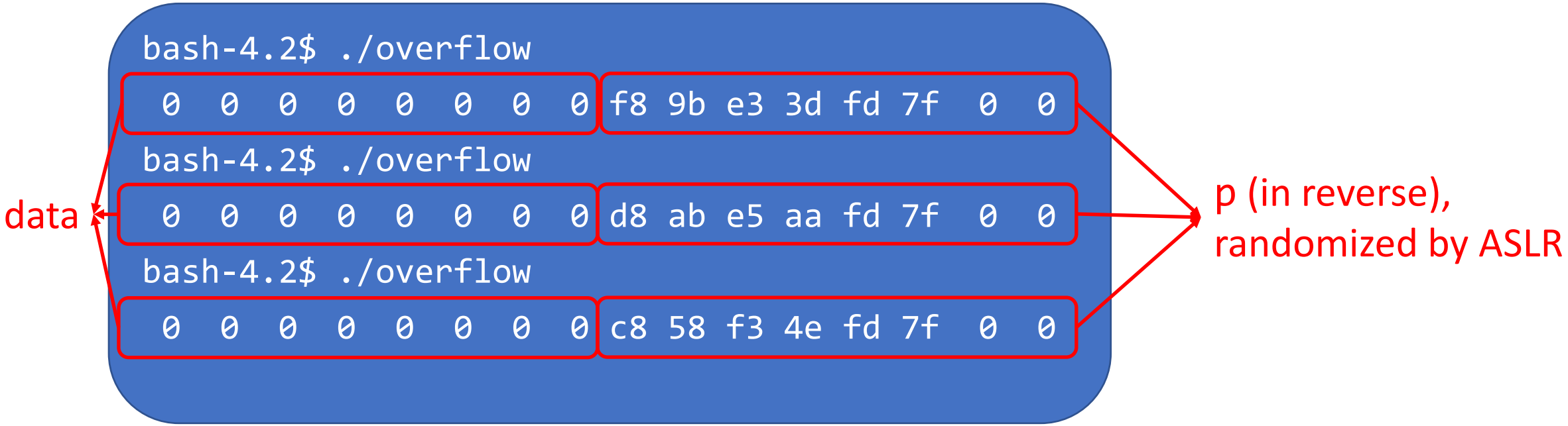
- Only happens in languages like C/C++ with direct access to memory
  - Does not happen in memory-managed languages like Java or Python
  - C/C++ is used to write most system code so still a big problem

Buffer Overflow Example

```
void send_data(char *data, int len) {
   for (int i=0; i < len; i++)
     printf("%2hhx ", data[i]);
   printf("\n");
}
int main() {
   char *p;
   char data[8] = {0};
   p = data;
   send_data(data, 16);
   return 0;
}
```

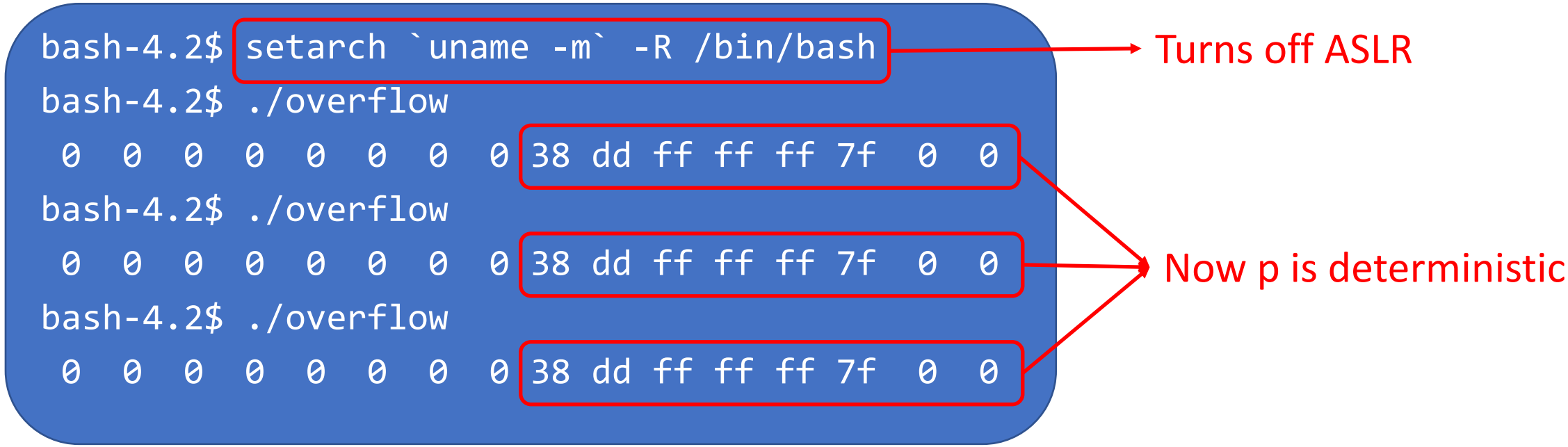


Buffer Overflow Output



- Randomized addresses can leak out to output due to a memory error!

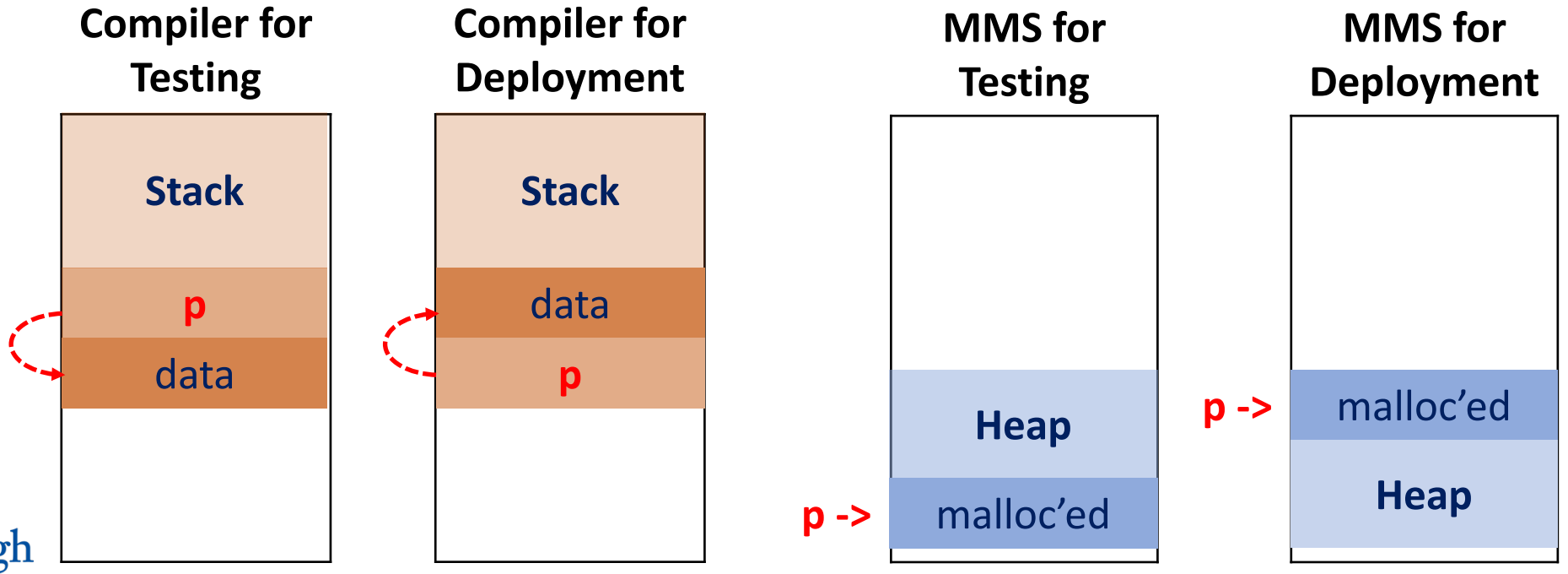
You could Turn Off ASLR …



- Can help in reproducing bugs in a debug setting (this is what GDB does)
- But clients will still want ASLR on for security → surprise defects can still happen

You could Turn Off ASLR …

- Even if client does not use ASLR, things can still go wrong
  - If binary deployed to client uses different *compiler* than test version (Or same compiler but different compile options)
  - If client uses a different runtime *memory management system* (MMS) (Or same MMS but MMS is nondeterministic on parallel mallocs)



What to do? Stamp Out the Error!

- Let's use Google Address Sanitizer for this purpose

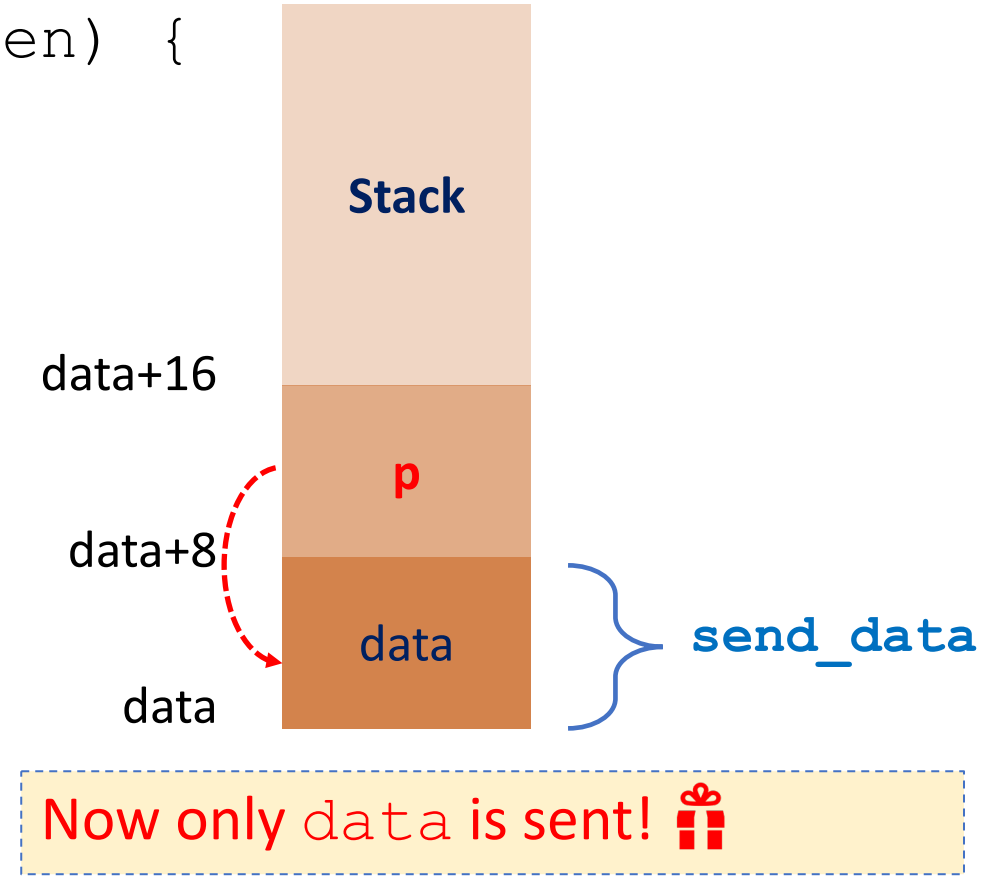
```
bash-4.2$ clang overflow.c -fsanitize=address -g –o overflow
bash-4.2$ ./overflow
==357==ERROR: AddressSanitizer: stack-buffer-overflow on …
READ of size 1 at 0x7fffffffdc88 thread T0
    #0 0x4f858c in send_data overflow.c:7:22
    #1 0x4f86f1 in main overflow.c:17:3
    …
```

Tells compiler to sanitize addresses

Where in code buffer overflow happened

Buffer Overflow Fixed

```
void send_data(char *data, int len) {
   for (int i=0; i < len; i++)
     printf("%2hhx ", data[i]);
   printf("\n");
}
int main() {
   char *p;
   char data[8] = {0};
   p = data;
   printf("p = %p\n", p);
   send_data(data, 8);
   return 0;
}
```



Google Address Sanitizer (ASAN)

- Part of Google sanitizer suite
  - Address Sanitizer (ASAN)
  - Thread Sanitizer (TSAN)
  - Memory Sanitizer (MSAN)
  - And more.
- Works through *instrumentation*
  - Inserting extra instructions into program for the purpose of monitoring
  - Instrumentation code reports back issues during/after execution of code
- Compilers where available (with the `-fsanitize=address` switch)
  - LLVM (clang) starting with version 3.1
  - GCC starting with version 4.8

ASAN Only Sees Errors that Happen

```
#include <stdio.h>
#include <stdlib.h>

void send_data(char *data, int len) {
  for (int i=0; i < len; i++)
    printf("%2hhx ", data[i]);
  printf("\n");
}
int main(int argc, char *argv[]) {
  char *p;
  char data[8] = {0};
  p = data;
  send_data(data, atoi(argv[1]));
  return 0;
}
```

```
bash-4.2$ ./overflow 8
 0  0  0  0  0  0  0  0
bash-4.2$ ./overflow 16
==517==ERROR: AddressSanitizer:
  stack-buffer-overflow on …
READ of size 1 at 0x7fa6c8700028
    #0 … in send_data overflow.c:6
    #1 … in main overflow.c:13
    …
```

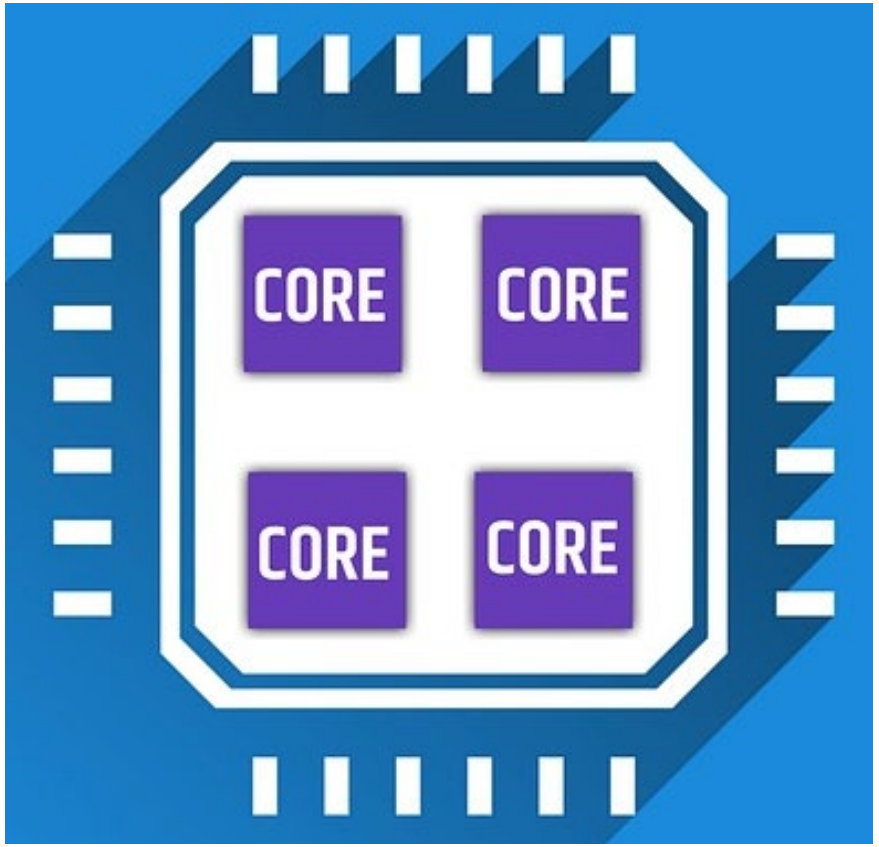
- ASAN only sees bugs observed during execution
- With argument 8, the bug goes unnoticed
- Tests must use inputs that trigger the bug

Outline

- Nondeterminism by mistake
  - Memory errors (examples / solutions)
  - Datarace errors (examples / solutions)
- Nondeterminism by design
  - Random number generation (examples / solutions)
  - Thread interleaving (examples / solutions)
- Summary

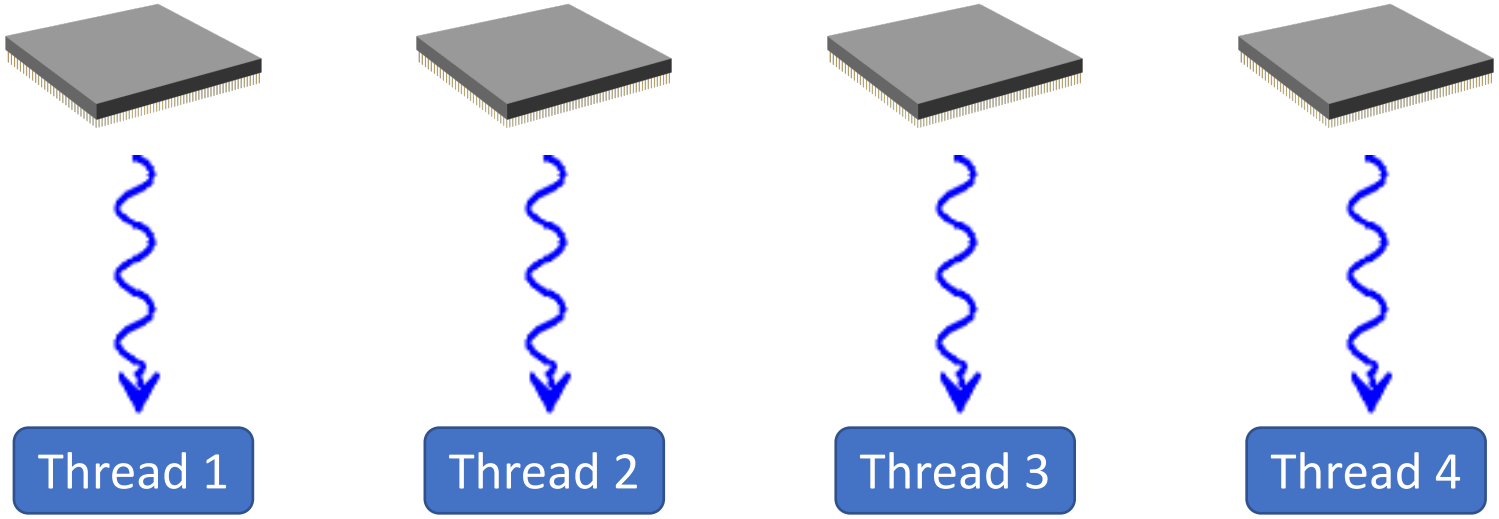
First, an Intro to Parallel Programming

- Your laptop or cellphone has multiple CPUs (Seen on the right: quad-core with 4 CPUs)
- A program runs on just 1 CPU by default (Using just 25% of the computing power!)
- A parallel program can use all 4 CPUs (Utilizing your computer 100%)



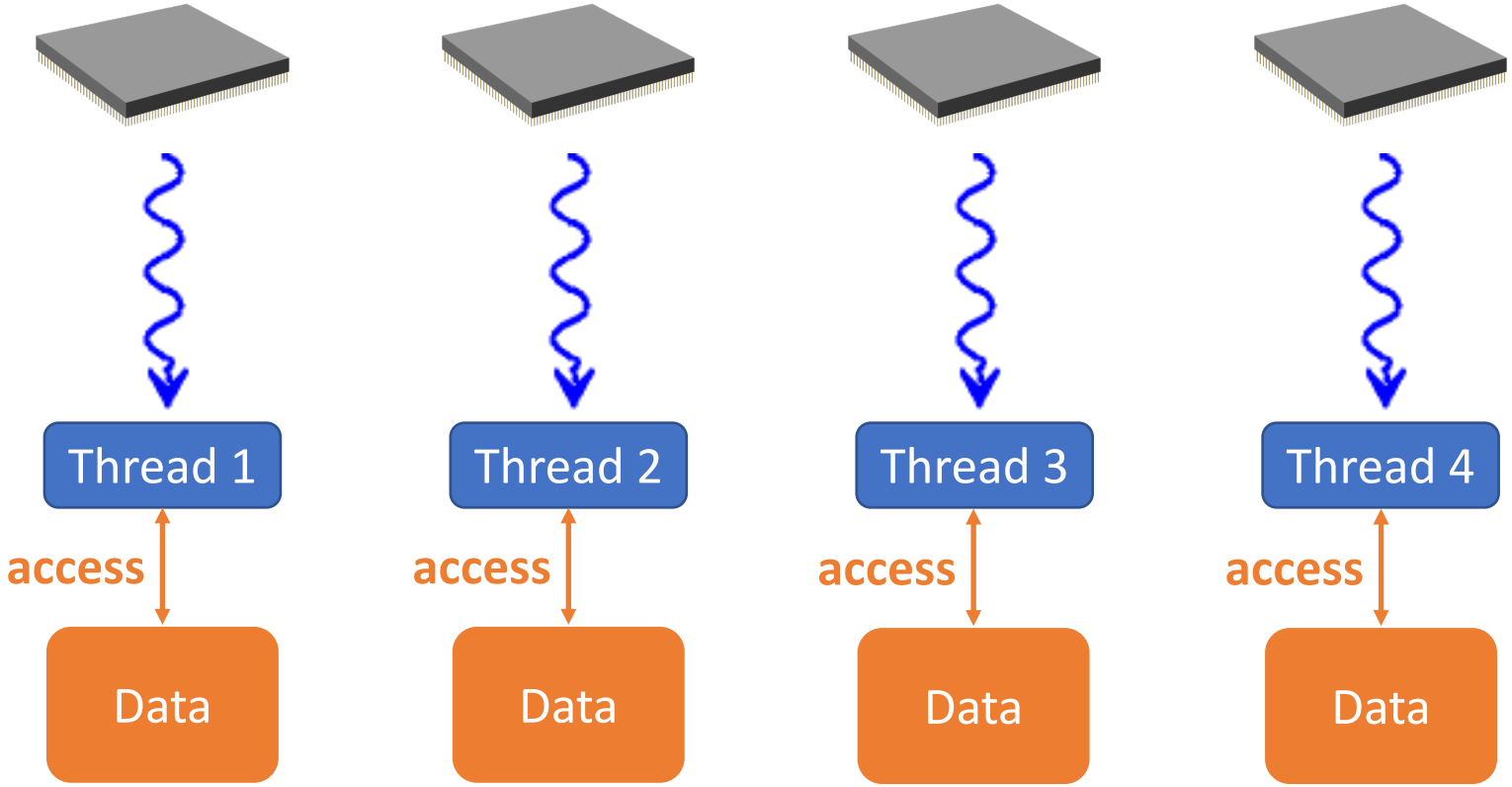
A Parallel Program Runs 1 Thread per CPU

- Thread: a unit of code that runs on 1 CPU



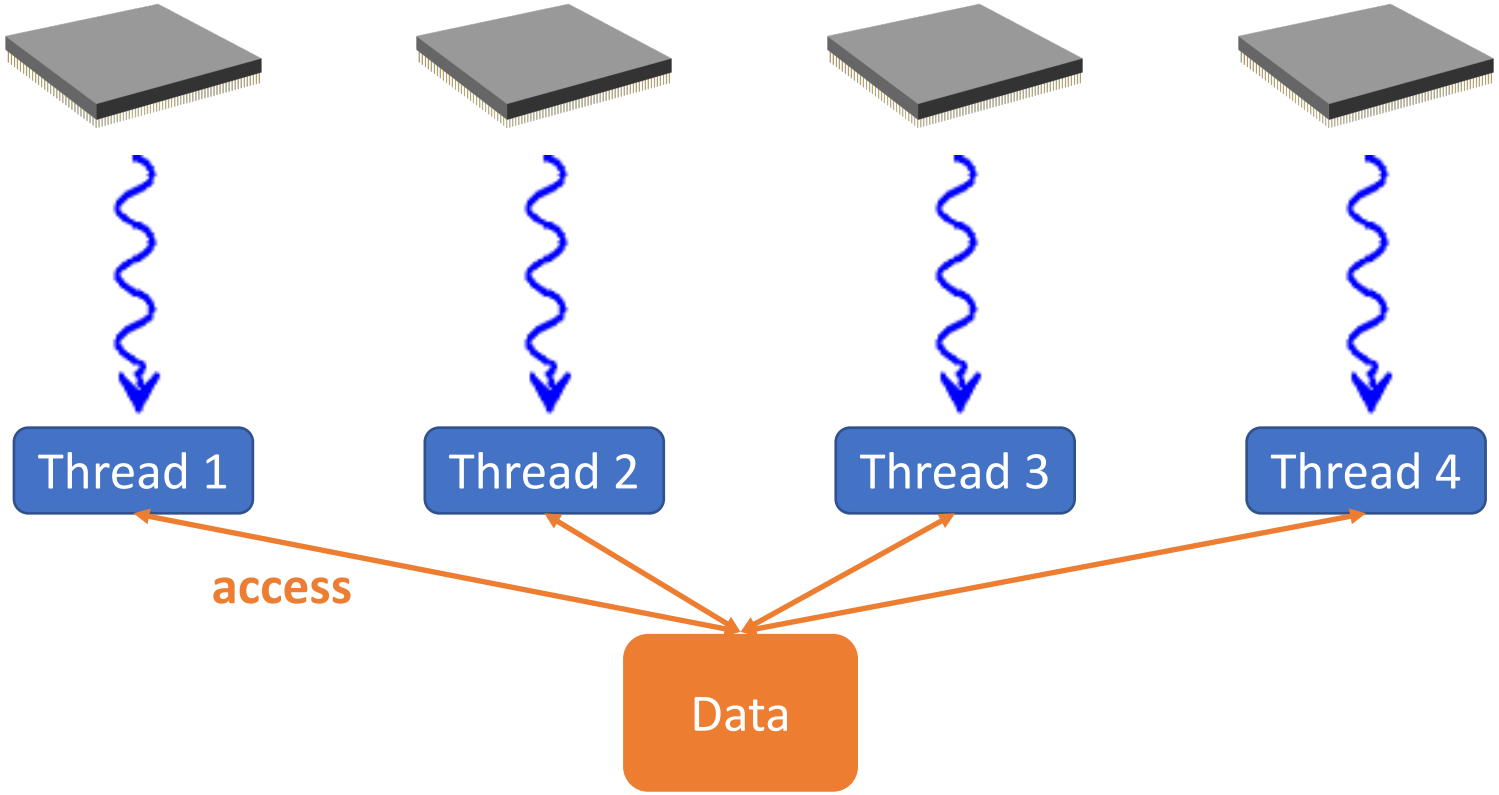
Mostly, a Thread Works on its Own Data

- Thread: a unit of code that runs on 1 CPU



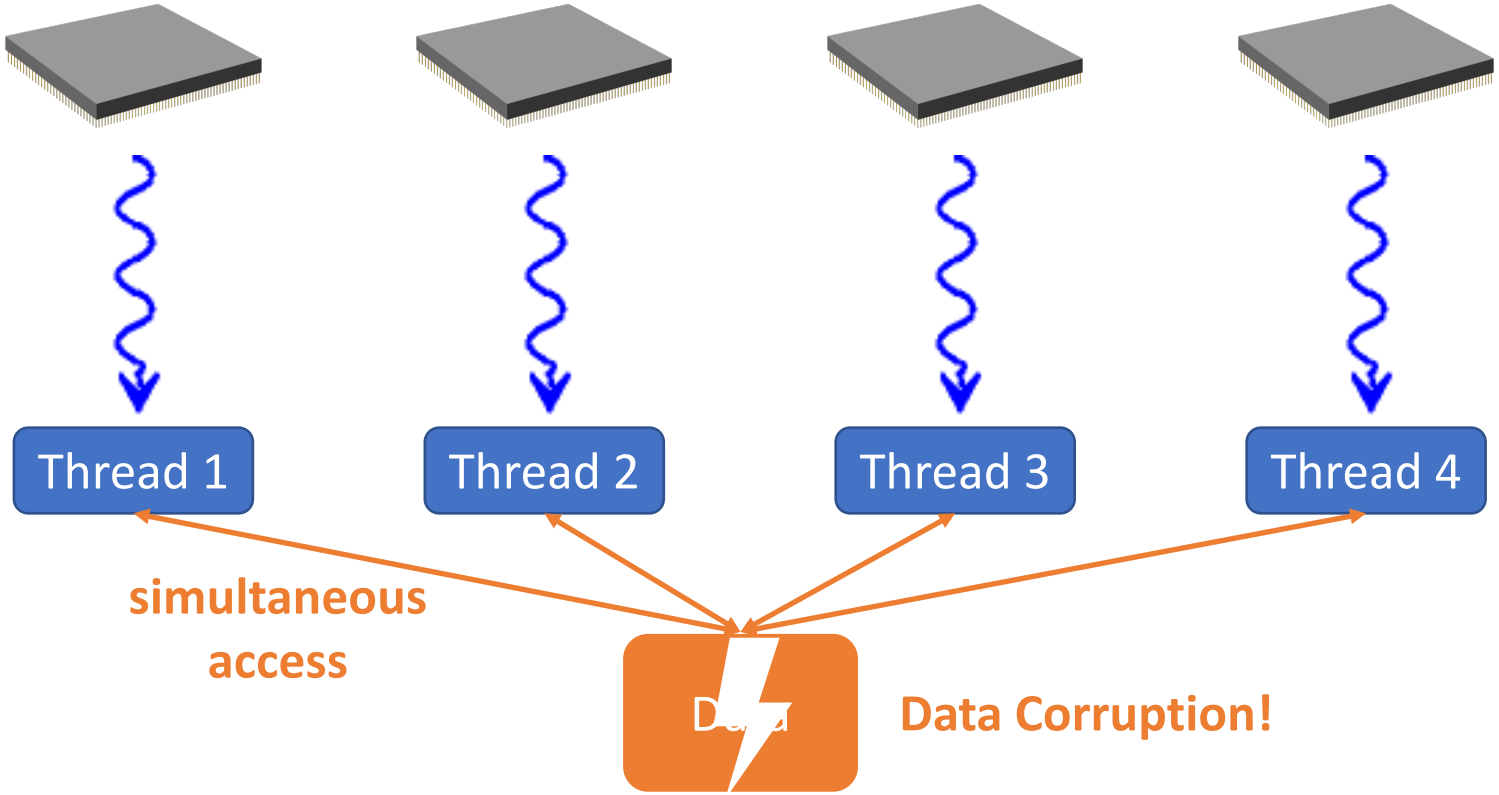
Sometimes, Threads Work on Shared Data

- Thread: a unit of code that runs on 1 CPU



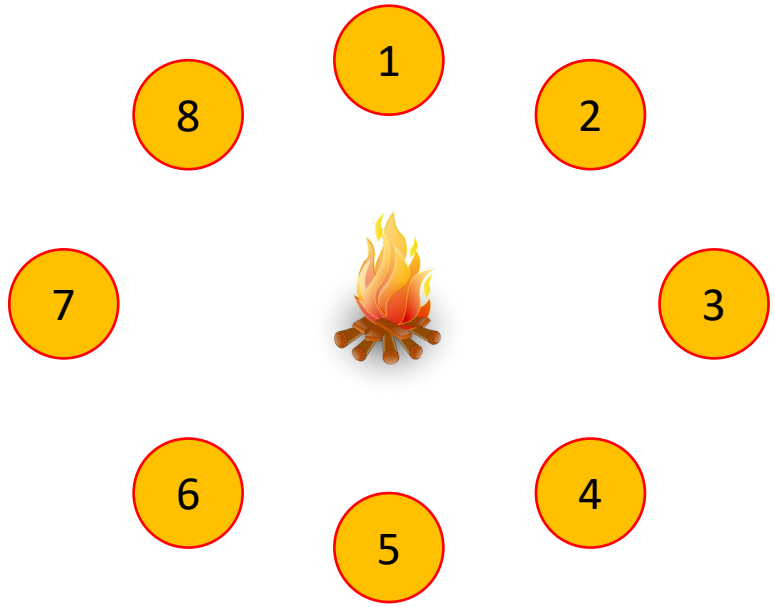
If not Careful, Data can be Corrupted!

- If threads don't take turns, and access data simultaneously



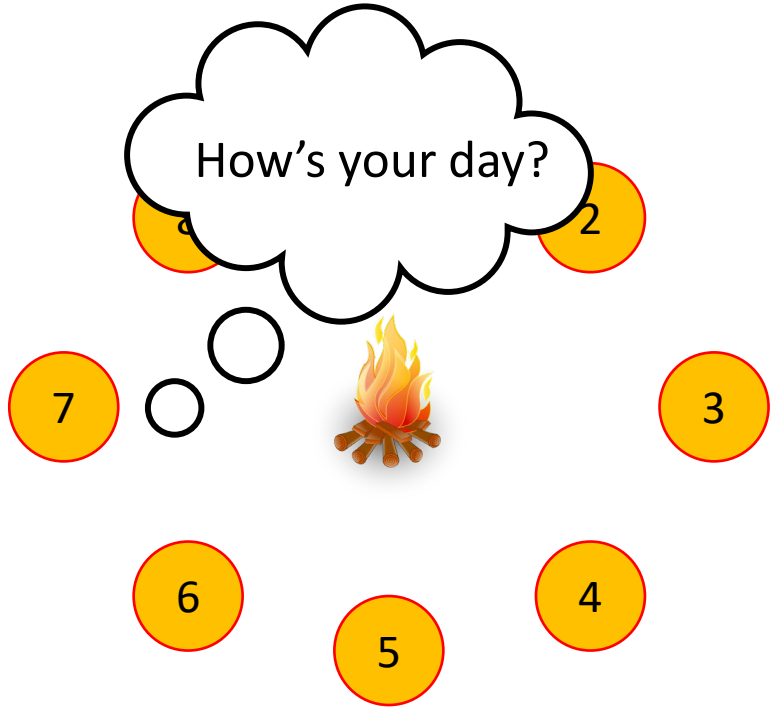
Why Data Corruption on Simultaneous Access?

- Imagine a group of village elders having a discussion around a campfire



Why Data Corruption on Simultaneous Access?

- If the elders speak one at a time, they will understand each other



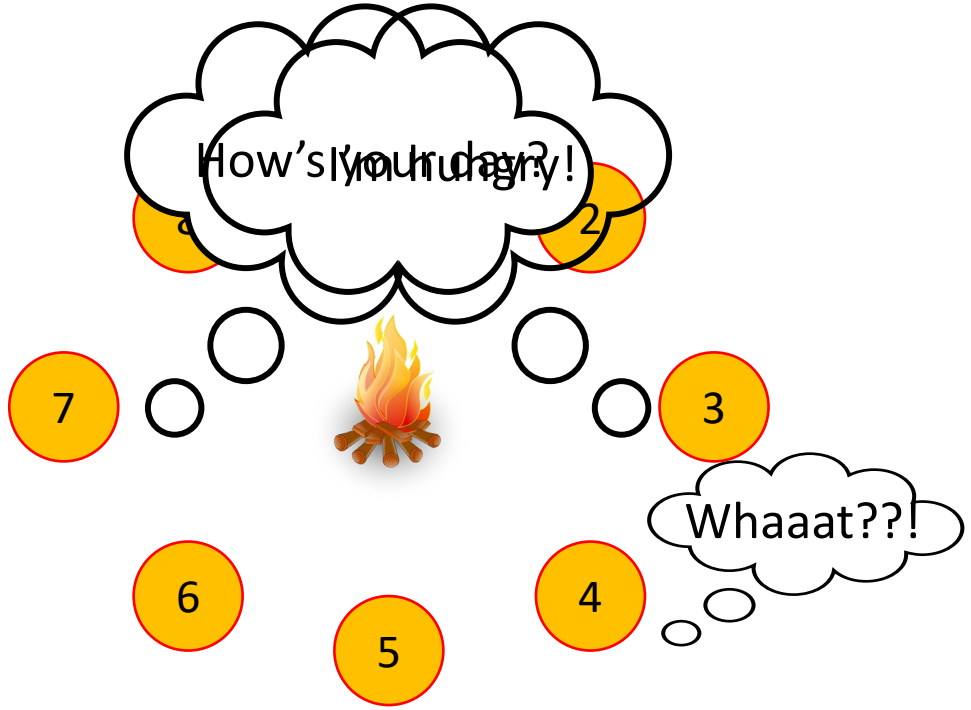
Why Data Corruption on Simultaneous Access?

- If the elders speak one at a time, they will understand each other



Why Data Corruption on Simultaneous Access?

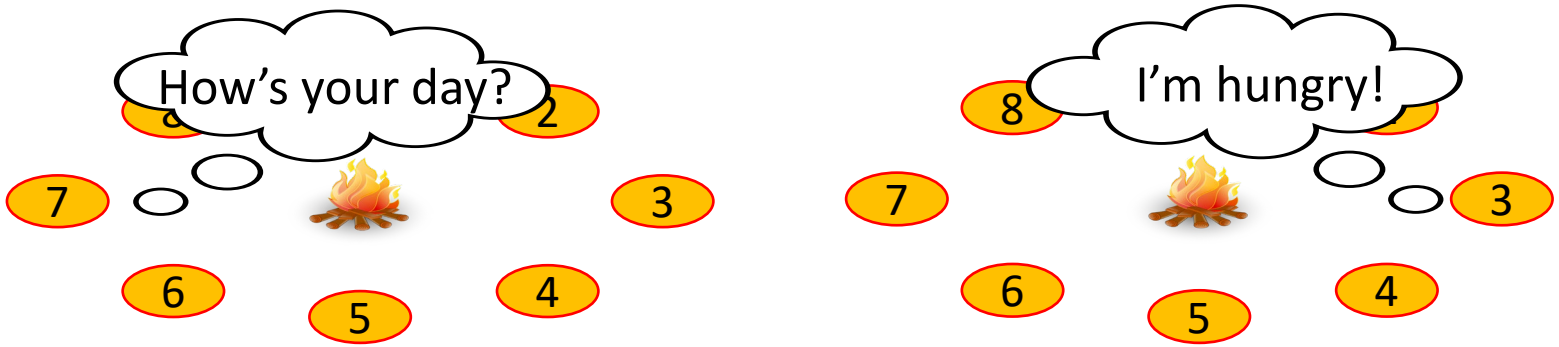
- But if they talk at the same time, the speech becomes garbled



- Same thing happens with data. This is called a *data race*.

Worst Part: Data Races are Nondeterministic

- If lucky, meeting may adjourn with everyone taking turns speaking



- If unlucky, two elders may speak at the same time causing the problem

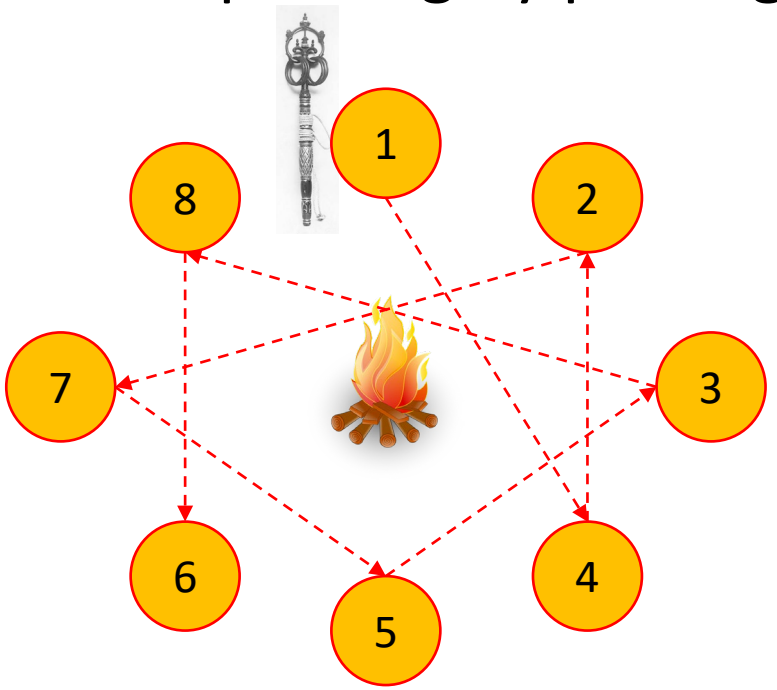


- It all depends on timing …

Solution: Speaking Staff



- Rule: only the elder with the speaking staff shall speak
- Forces elders to take turns speaking by passing around the staff



Speaking Staff in Software is the *Lock*

- *Lock*: a software object that only one thread can hold at a time
- Threads take turn accessing shared data by passing around a lock
- Data races can be removed with proper use of locks

Datarace Example

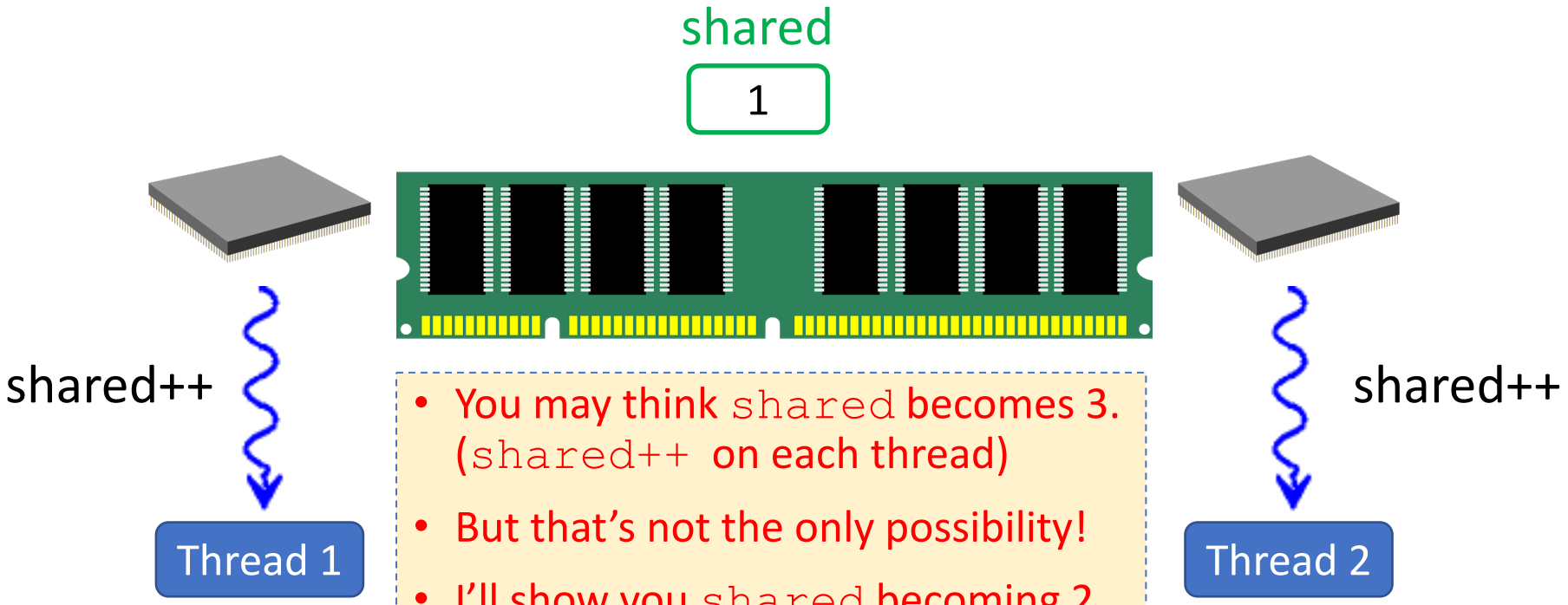
```
int shared = 0;
void *add(void *unused) {
   for(int i=0; i < 1000000; i++) { shared++; }
   return NULL;
}
int main() {
   pthread_t t;
   // Child thread starts running add
   pthread_create(&t, NULL, add, NULL);
   // Main thread starts running add
   add(NULL);
   pthread_join(t, NULL);
   printf("shared=%d\n", shared);
   return 0;
}
```

```
bash-4.2$ ./datarace
shared=1085894
bash-4.2$ ./datarace
shared=1101173
bash-4.2$ ./datarace
shared=1065494
```

- What do you expect from running this?
- Maybe shared=2000000 ?
- Due to datarace on `shared`.

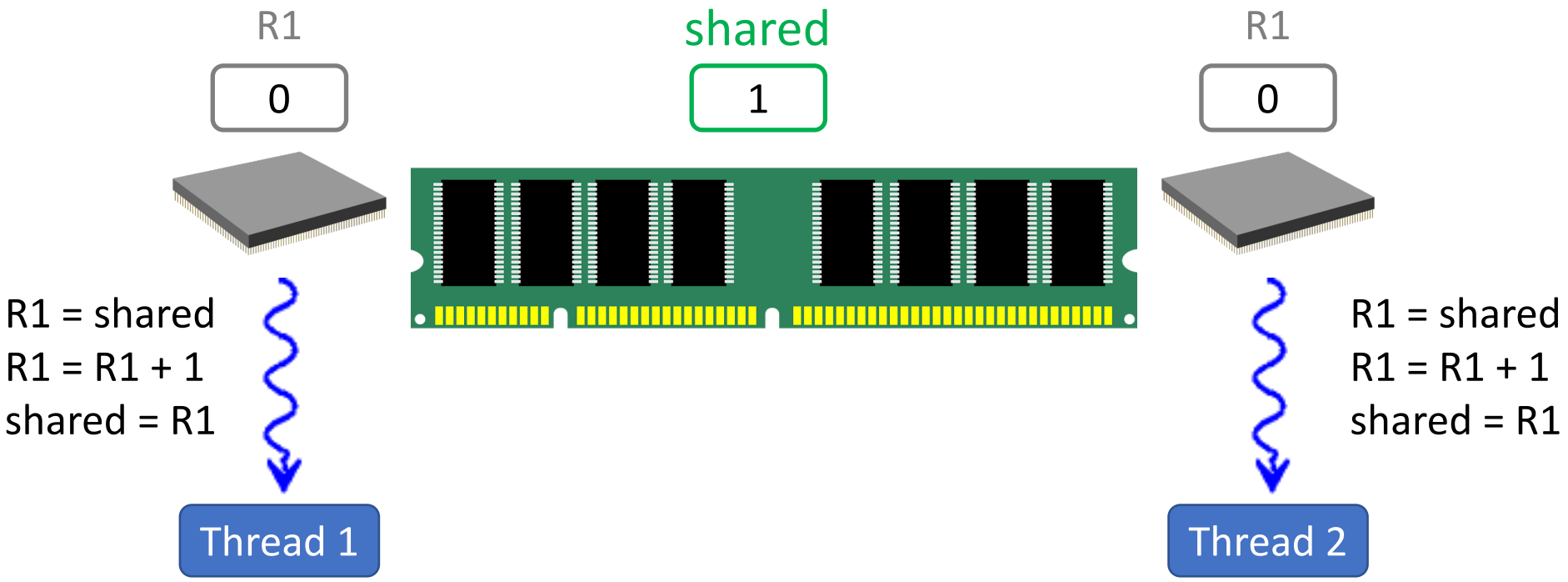
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



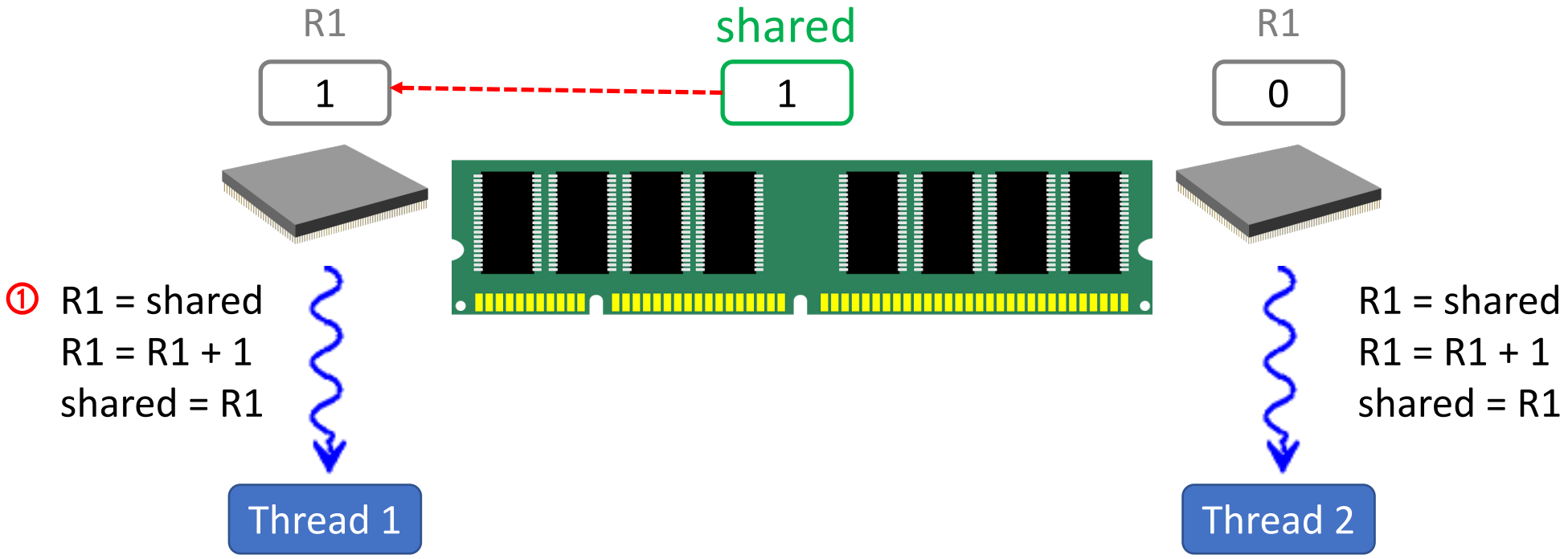
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



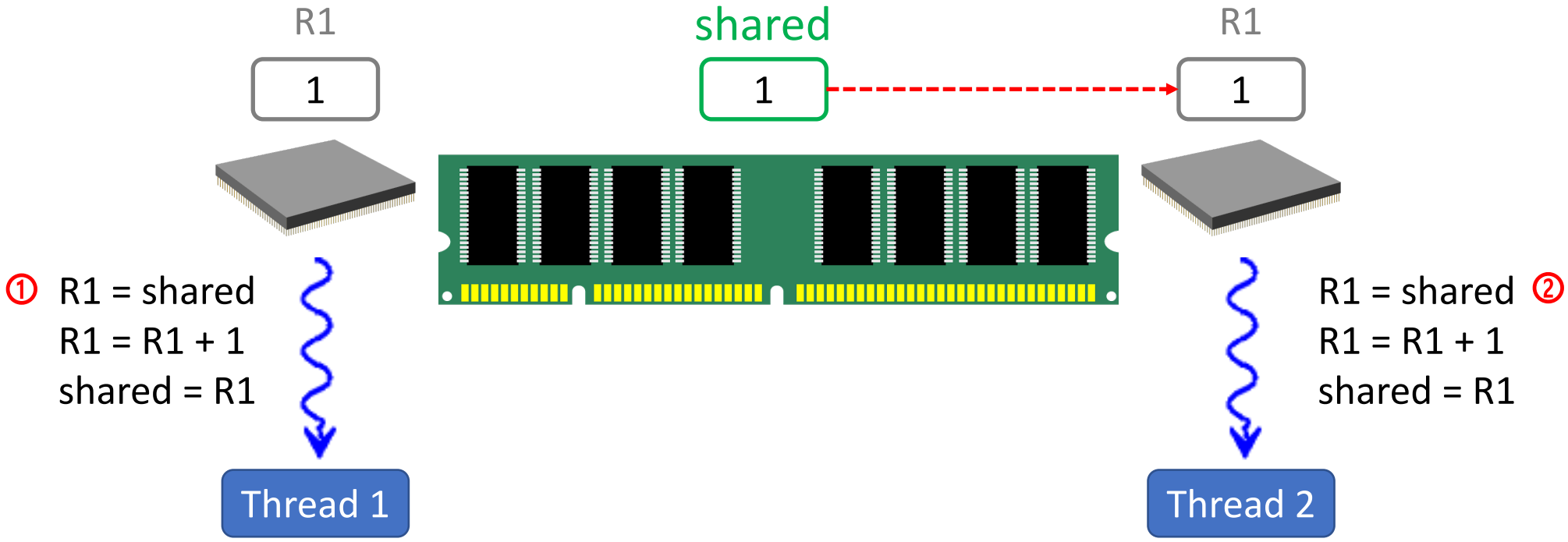
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



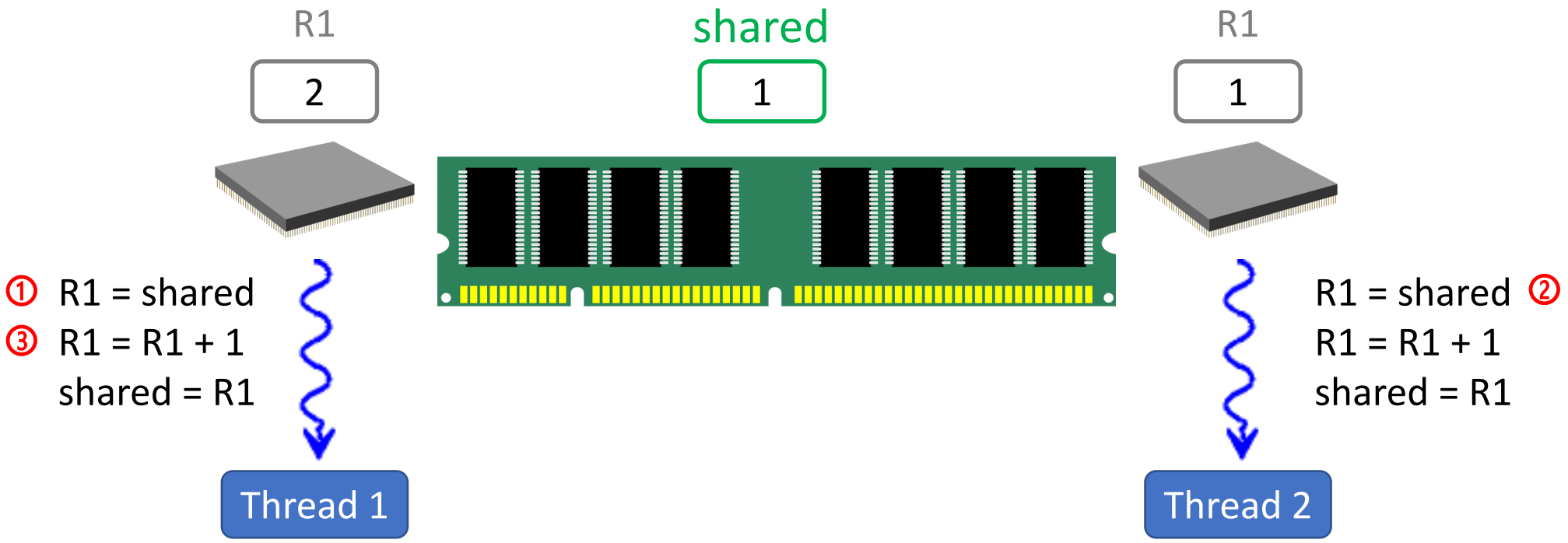
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



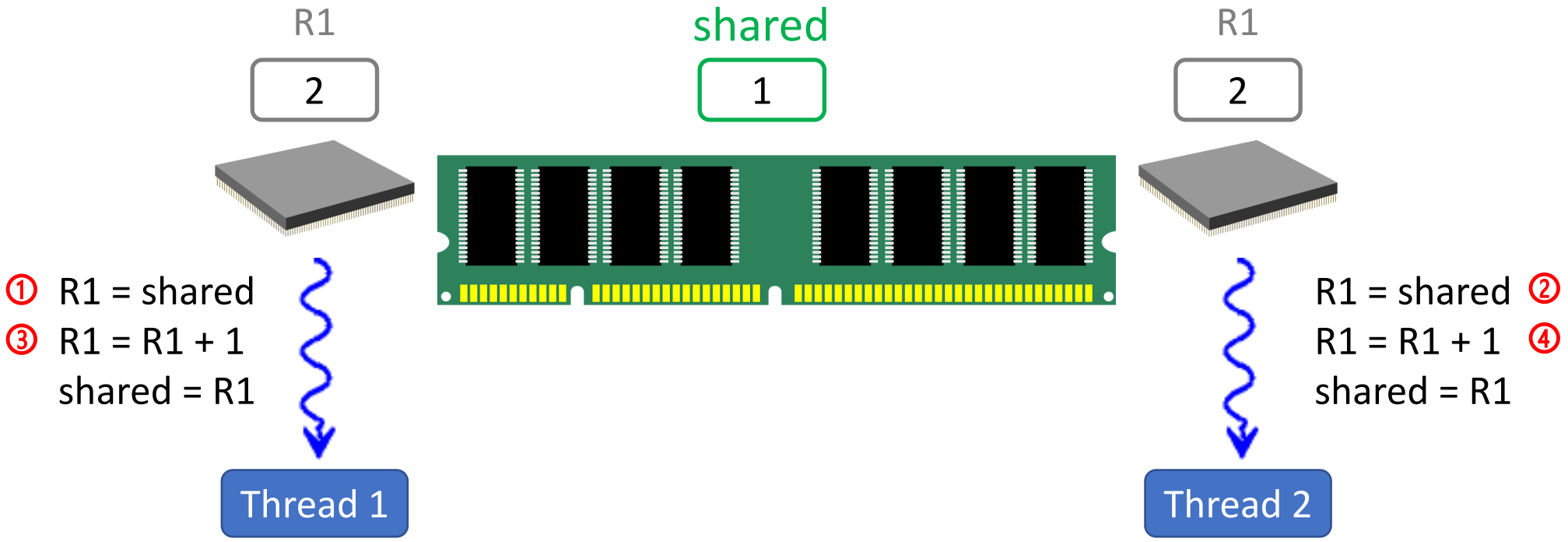
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



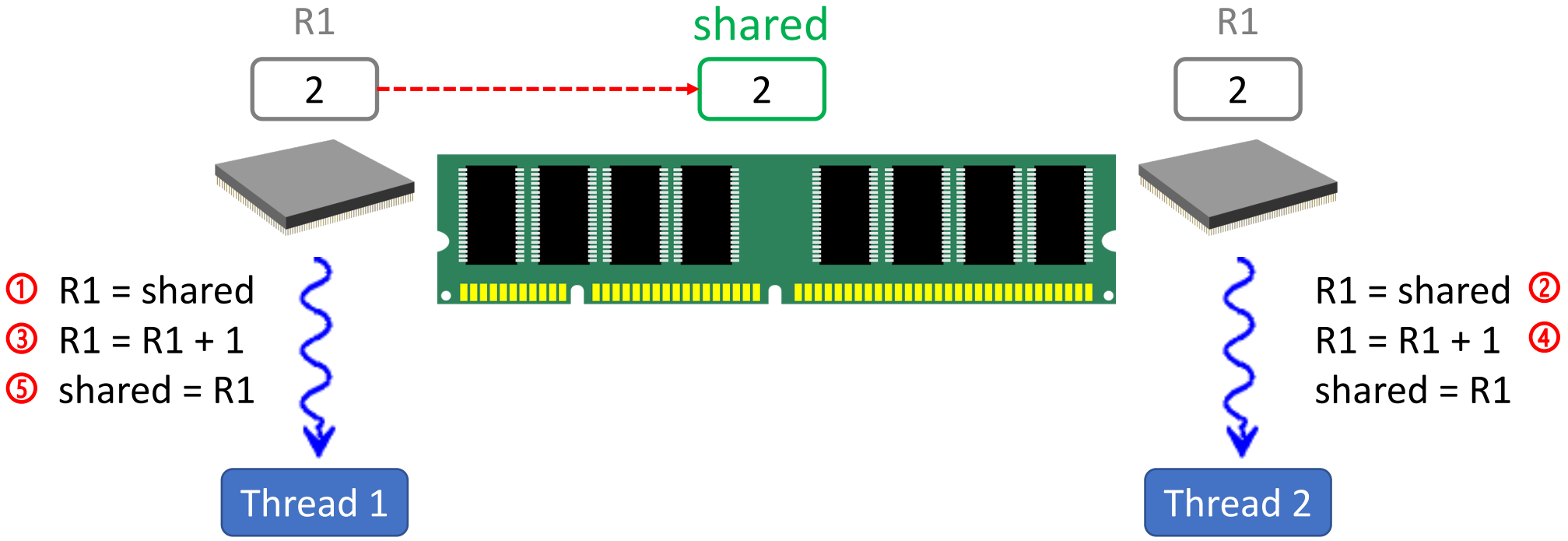
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



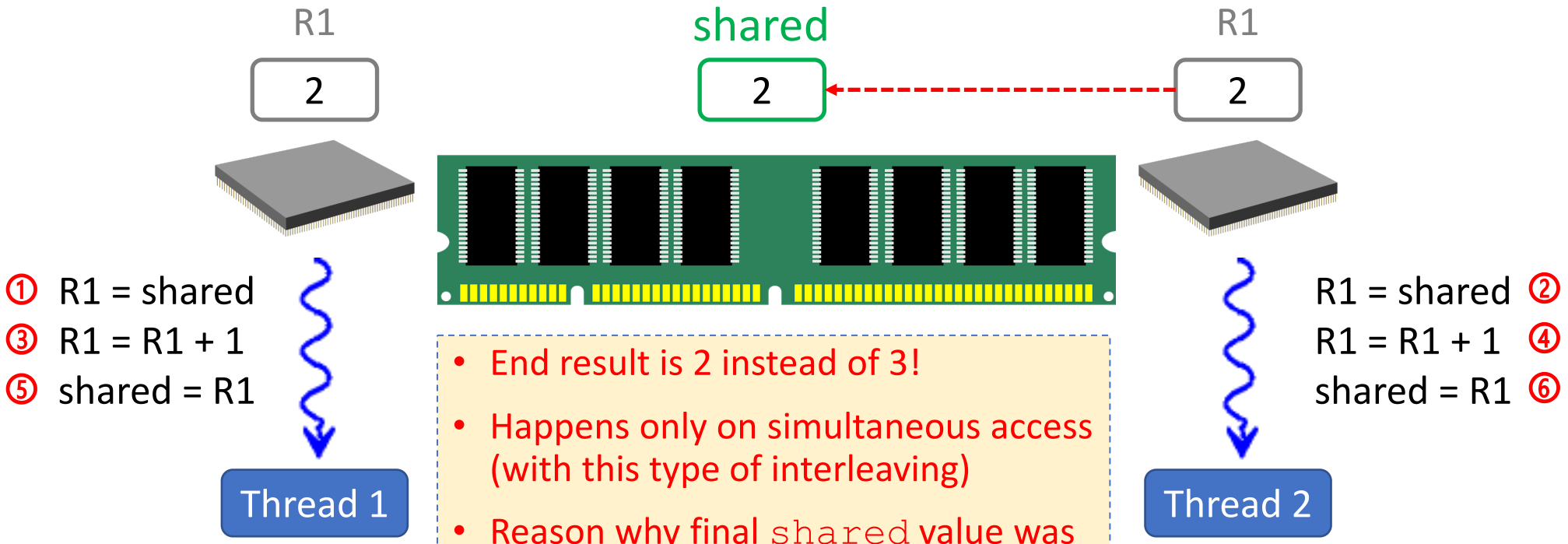
Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



Datarace: Sequence of Events

- When two threads do `shared++;` initially `shared` = 1



What to do? Stamp Out the Error!

- Let's use Google Thread Sanitizer this time!

```
bash-4.2$ clang datarace.c -fsanitize=thread -g -o datarace
bash-4.2$ ./datarace
WARNING: ThreadSanitizer: data race (pid=14291)
  Write of size 4 at 0x00000112d618 by main thread:
    #0 add datarace.c:5:42 (datarace+0x4ca832)
    #1 main datarace.c:11:3 (datarace+0x4ca89f)

  Previous write of size 4 at 0x00000112d618 by thread T1:
    #0 add datarace.c:5:42 (datarace+0x4ca832)
  …
```

Tells compiler to sanitize threads

Where in code dataraces happened

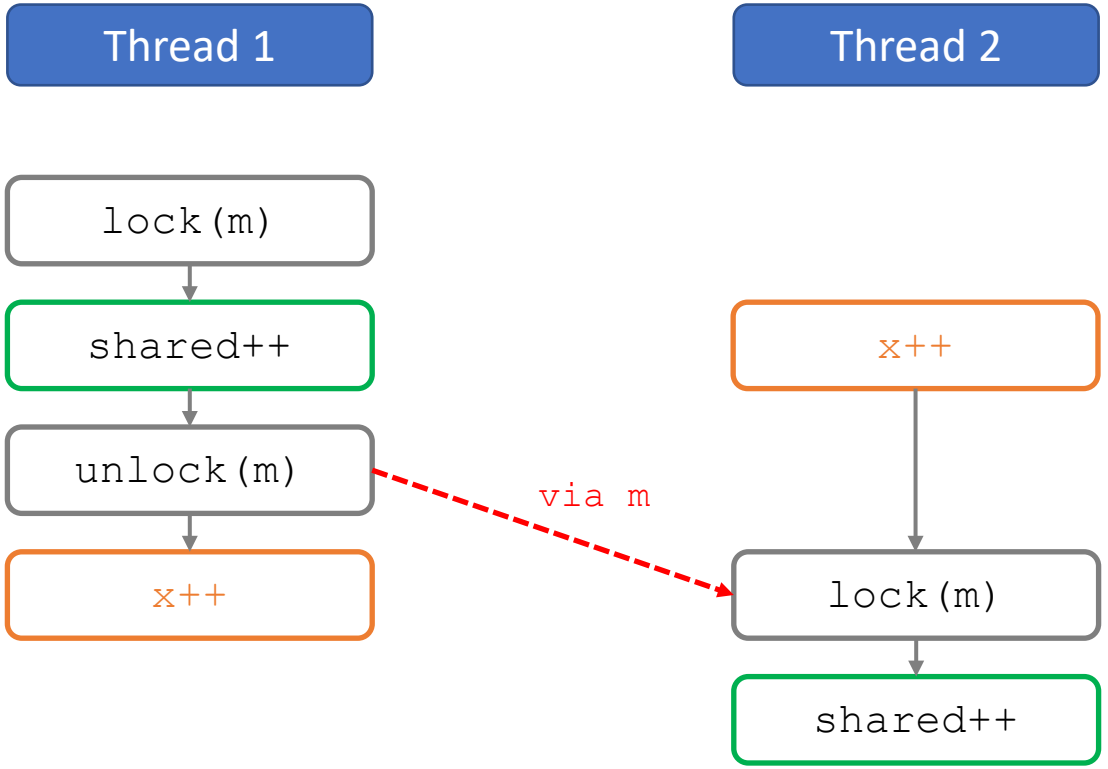
Datarace Fixed

```
pthread_mutex_t lock;
int shared = 0;
void *add(void *unused) {
  for(int i=0; i < 1000000; i++) {
      pthread_mutex_lock(&lock);
      shared++;
      pthread_mutex_unlock(&lock);
  }
  return NULL;
}
int main() {
  …
}
```

```
bash-4.2$ ./datarace
shared=2000000
bash-4.2$ ./datarace
shared=2000000
bash-4.2$ ./datarace
shared=2000000
```

- Now result is deterministic
- Threads take turns accessing `shared`

Happens-Before: Ordering Accesses

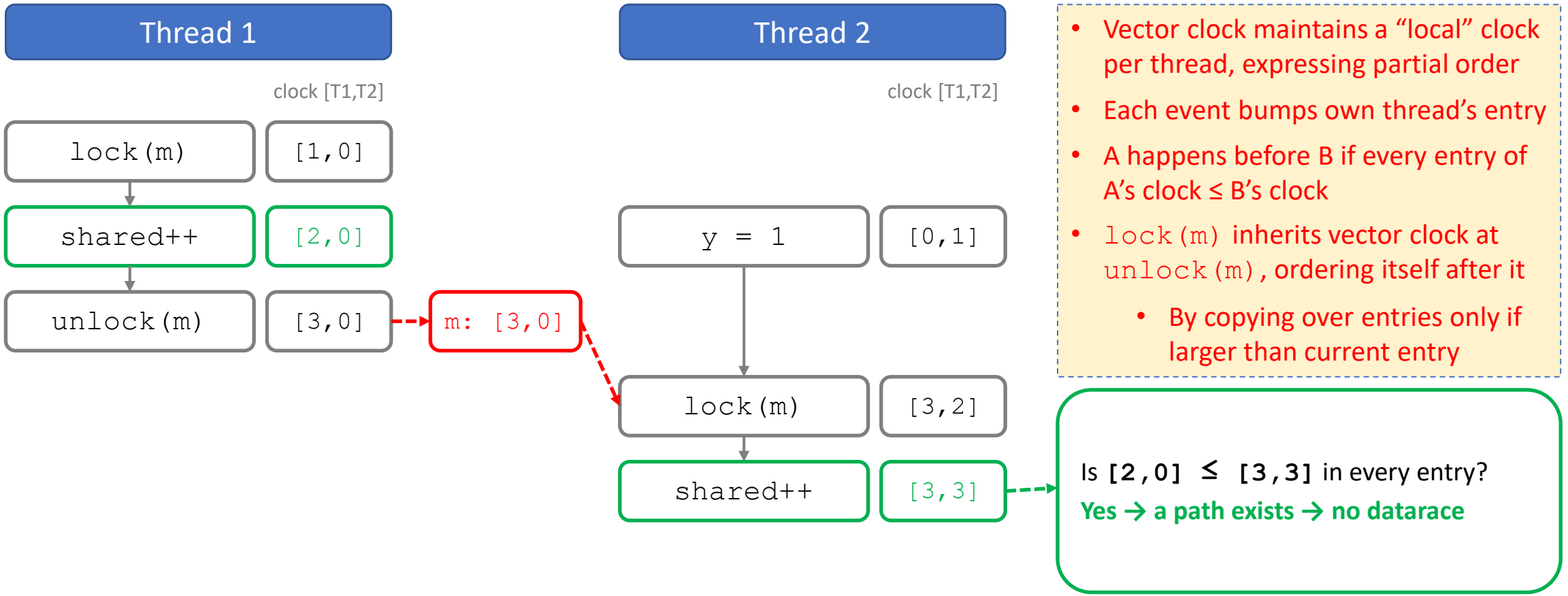


- A happens before B if a path of arrows leads from A to B
- Arrows:
  - Program order in a thread
  - `unlock(m)` → next `lock(m)`
- Happens-before is a partial order:
  - Some pairs are unordered (accesses to `x++`)
- Datarace: two unordered accesses to the same variable, where at least one is a write
  - Why? It means they can happen concurrently or far apart depending on timing.

✓ A path leads from T1's `shared++` to T2's `shared++`: ordered, no datarace

✗ No path between T1's `x++` and T2's `x++`: unordered, datarace

TSAN Uses Vector Clocks to Track Partial Order



- Vector clock maintains a "local" clock per thread, expressing partial order
- Each event bumps own thread's entry
- A happens before B if every entry of A's clock ≤ B's clock
- `lock(m)` inherits vector clock at `unlock(m)`, ordering itself after it
  - By copying over entries only if larger than current entry

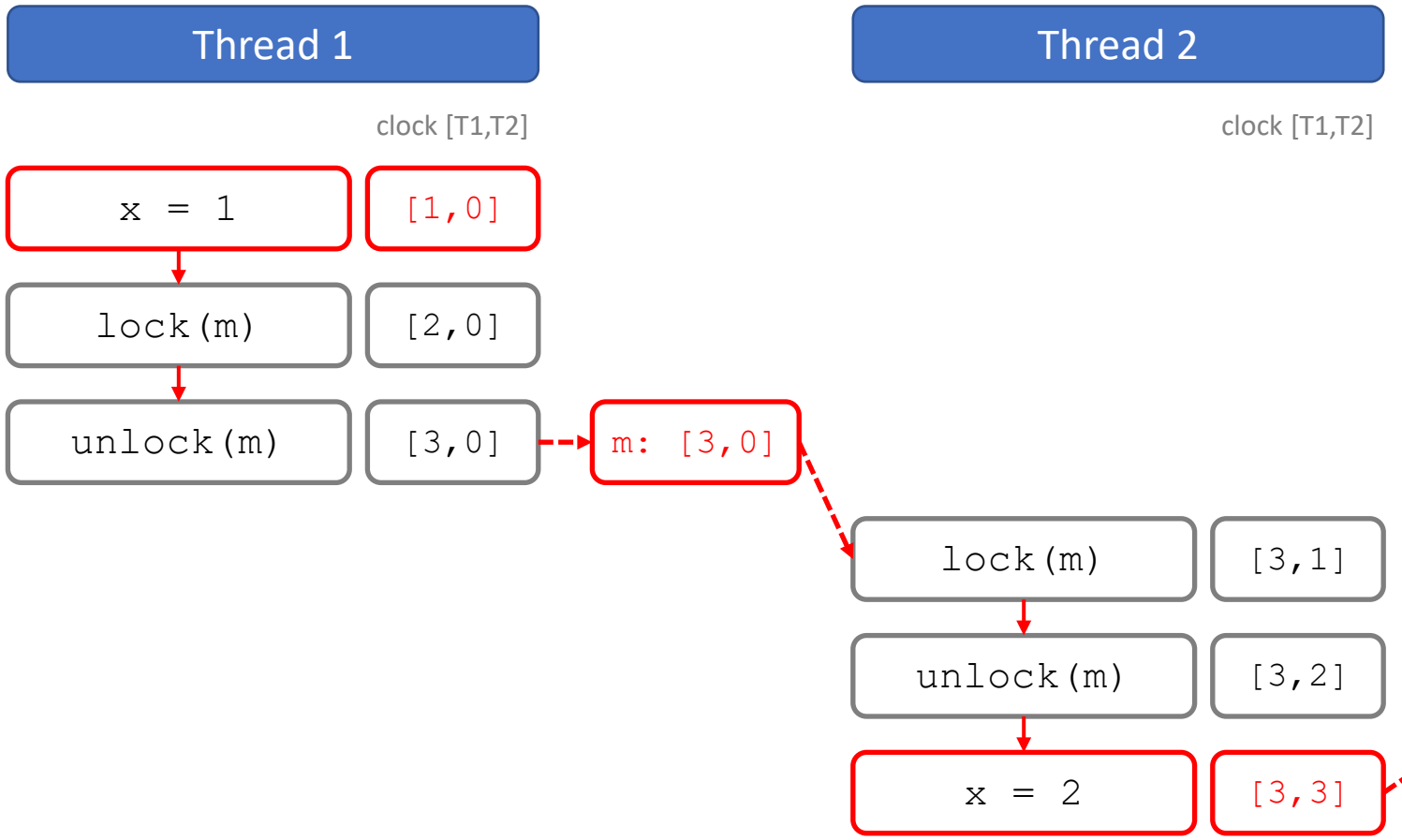
TSAN Only Sees Races that Happen

```
int x = 0;
pthread_mutex_t m = PTHREAD_MUTEX_INITIALIZER;
void *T1(void *unused) {
  x = 1;
  pthread_mutex_lock(&m);
  pthread_mutex_unlock(&m);
  return NULL;
}
void *T2(void *unused) {
  pthread_mutex_lock(&m);
  pthread_mutex_unlock(&m);
  x = 2;
  return NULL;
}
int main() {
  … // run T1 and T2 in parallel, join
  printf("x = %d\n", x);
}
```

```
bash-4.2$ ./race
x = 2
bash-4.2$ ./race
x = 2
bash-4.2$ ./race
WARNING: ThreadSanitizer: data race
  Write of size 4 … by thread T1:
    #0 T1 race.c
  Previous write … by thread T2:
    #0 T2 race.c
x = 1
```

- TSAN only sees dataraces observed during execution
- It depends on which thread locks `m` first

Why TSAN Missed the Datarace



- In this run, `m` creates a path from `x = 1` to `x = 2`
- But `m` never protected `x`
- If T2 locked `m` first, there is no path: TSAN would have reported the race

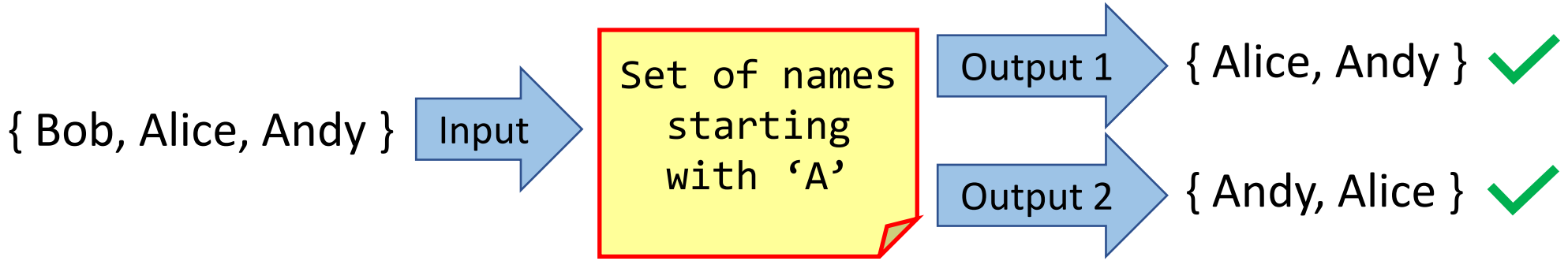
Is **[1,0] ≤ [3,3]** in every entry?

Yes → a path exists → no report

Nondeterminism by Design

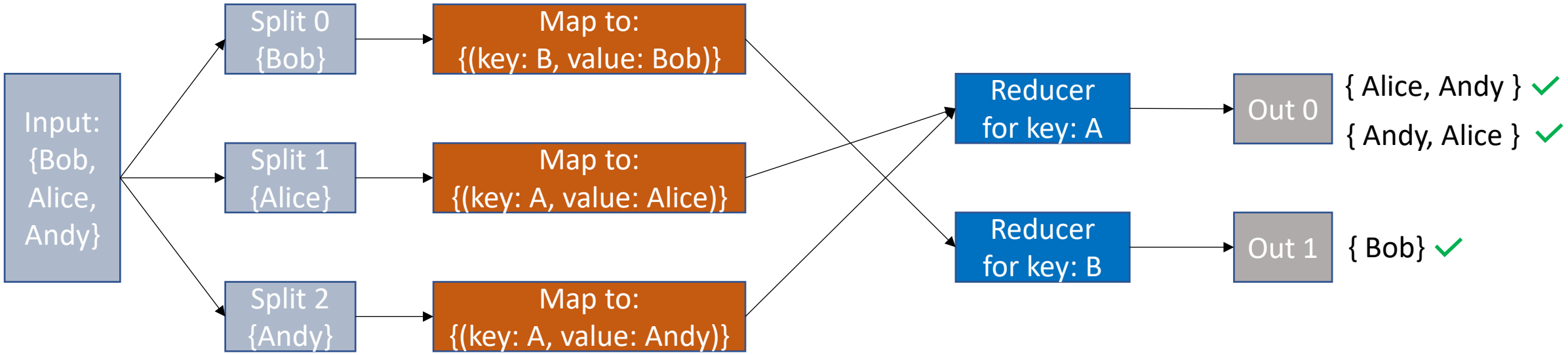
Why Nondeterminism by Design?

- Consider the following nondeterministic program:



  - Both outputs are correct since there is no ordering constraint in a set
- Less constraints usually means the program can run faster!
  - A straightforward loop checking each name is deterministic but slow
  - But what if we used parallel MapReduce to speed up the program?

MapReduce Implementation of Filter Names



Map:
String → (first letter of String, String)

Reduce:
Strings → concatenated CSV

- Reducer concatenates in the order of arrival → nondeterministic output!
- All non-commutative reducers like concatenation have this property
- For determinism, must constrain order of mapping → slows down program!

Nondeterminism is Inherent in Parallelism

- "Nondeterminism in MapReduce considered harmful? An empirical study on non-commutative aggregators in MapReduce programs."
  - Xiao, Tian et al., International Conference on Software Engineering, 2014.
  - https://www.microsoft.com/en-us/research/wp-content/uploads/2016/02/icsecomp14seip-seipid15-p.pdf

- Most other parallel frameworks suffer from nondeterminism as well
  - Including POSIX Threads, Java Threads, OpenMP, MPI, CUDA, TensorFlow
  - To allow flexibility in execution order for maximal performance

Why Nondeterminism by Design?

1. To make programs go faster through parallel execution
   - Sometimes all outputs are equally correct → nondeterminism is not a problem
   - Sometimes, for optimization problems, some outputs are "better" than others
     - Still, nondeterministic output is okay as long as it is better than deterministic output (given the same amount of time to run; i.e. nondeterministic approaches optimum faster)
     - True for NP-hard problems (e.g. integer linear programming, constraint solving, …)

2. To intentionally introduce randomness (random number generation)
   - Video games – to introduce random events into the game
   - Cryptography – to make cryptographic keys unpredictable

It’s by Design – Deal with it!

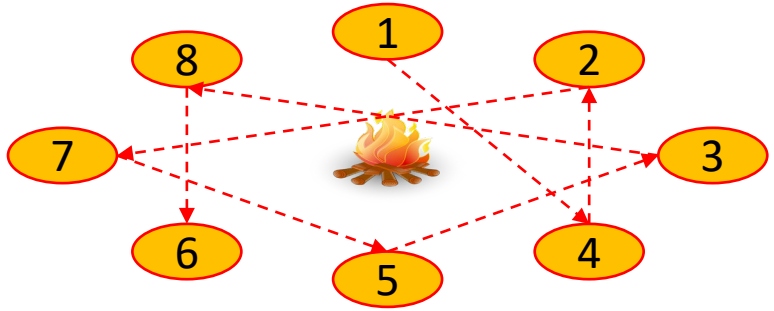
- Due to randomness by design
  - Random number generation
  - Thread interleaving
- You can’t just stamp out the nondeterminism. It’s by design.
- Somehow deal with the nondeterminism such that
  - You do not get any surprise defects at the client site
  - Defects are reproducible while debugging

Outline

- Nondeterminism by mistake
  - Memory errors (examples / solutions)
  - Datarace errors (examples / solutions)
- Nondeterminism by design
  - Thread interleaving (examples / solutions)
  - Random number generation (examples / solutions)
- Summary

Speaking Staff doesn't Remove All Nondeterminism

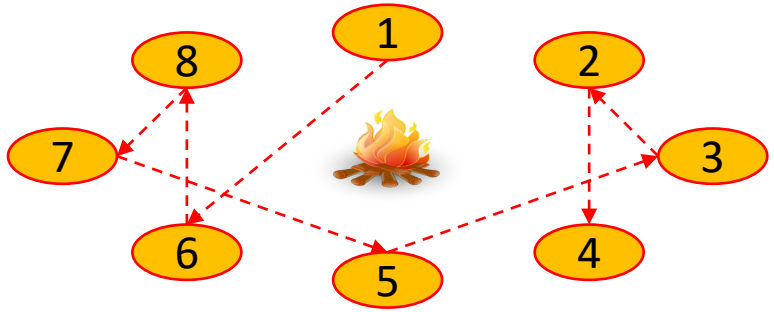
- Depending on the order the staff is passed, meeting script can change
- Order 1:



Meeting Script 1:



- Order 2:



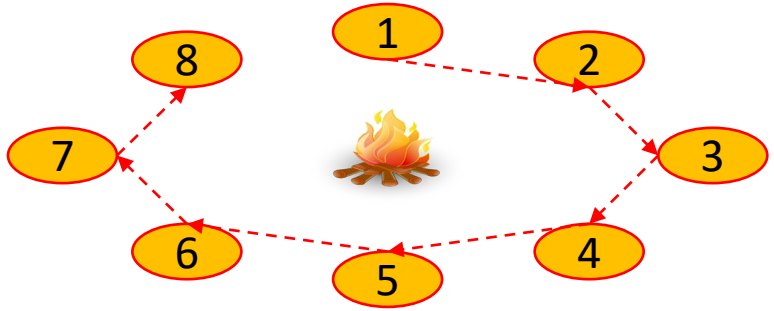
Meeting Script 2:



University of Pittsburgh

For Full Determinism, Must Fix Passing Order

- For example, fix staff passing order to clockwise direction
- Fixed clockwise order:



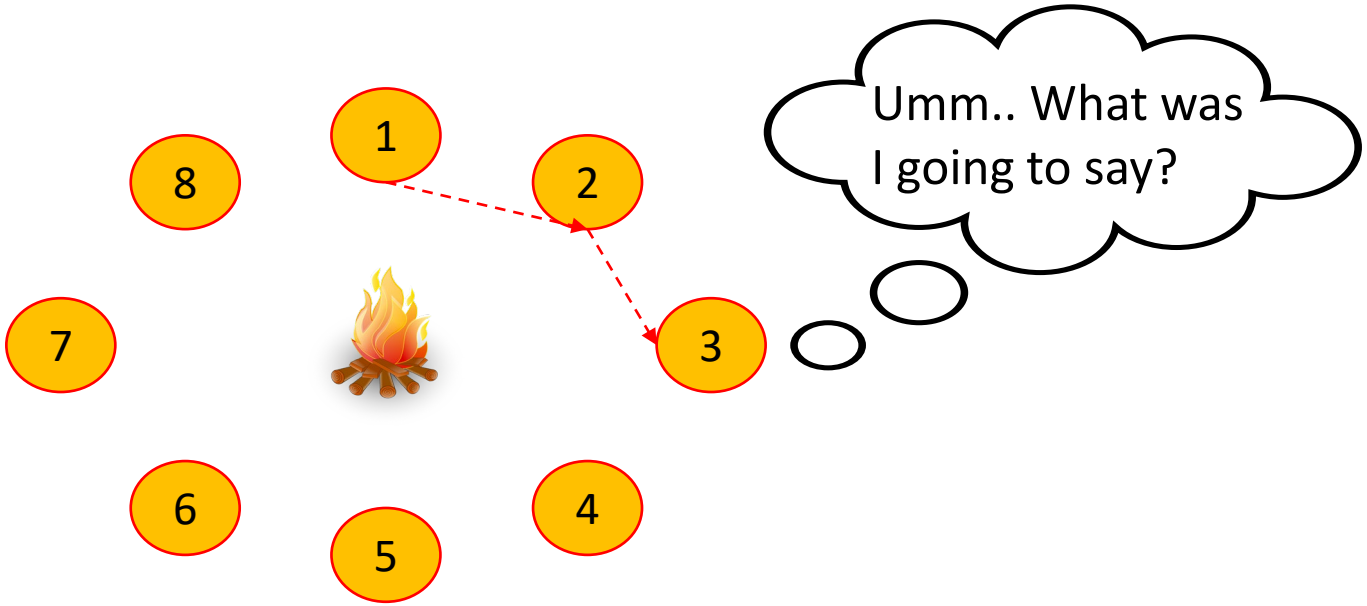
Fixed meeting script



- But programmers don't like doing this because it hurts performance

Performance Slow Down due to Fixed Order

- If an elder is not ready to speak, it can slow down the entire meeting



- Doesn't happen if staff is passed nondeterministically on demand

→ Many programs don't constrain thread interleaving for this reason

Nondeterministic Interleaving Example

```
class Interleaving implements Runnable {
  public static String script = "";
  public void run() {
    synchronized(this) {
      script += Thread.currentThread().getName() + " ";
    }
    synchronized(this) {
      script += Thread.currentThread().getName() + " ";
    }
  }

  public static void main(String[] args) throws InterruptedException {
    Interleaving m = new Interleaving();
    Thread t = new Thread(m);

    t.start();  // Child thread does run()
    m.run();    // Main thread does run()

    t.join();
    System.out.println(script);
  }
```

Java version of a lock, so no datarace.

But still nondeterministic due to interleaving.

- Main thread appends “main” twice
- Child thread appends “Thread-1” twice
- What are all the possible outputs?

University of Pittsburgh

Nondeterministic Interleaving Output

- 6 different possible outcomes!

```
bash-4.2$ java Interleaving
main main Thread-1 Thread-1

bash-4.2$ java Interleaving
main Thread-1 main Thread-1

bash-4.2$ java Interleaving
main Thread-1 Thread-1 main
```

```
bash-4.2$ java Interleaving
Thread-1 main main Thread-1

bash-4.2$ java Interleaving
Thread-1 main Thread-1 main

bash-4.2$ java Interleaving
Thread-1 Thread-1 main main
```

- All could be correct – if you don't care about the ordering. Or not.

Nondeterministic Interleaving is Problematic

- A defect may show up only on a particular interleaving
- No guarantee which interleaving will be chosen at runtime
- So testing becomes unreliable and unreproducible
- What to do? We'll get to it soon, but let's first talk about …

Outline

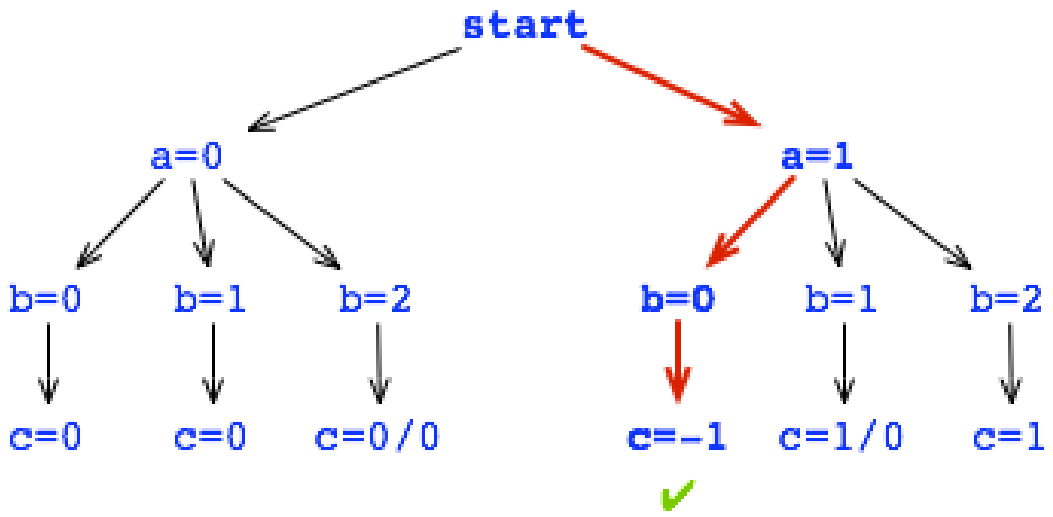
- Nondeterminism by mistake
  - Memory errors (examples / solutions)
  - Datarace errors (examples / solutions)
- Nondeterminism by design
  - Thread interleaving (examples / solutions)
  - Random number generation (examples / solutions)
- Summary

Random Number Generation Example

Given this code:

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```

If unlucky, paths with defects will not be covered during testing and bug may never be found!



① Random random = new Random()

② int a = random.nextInt(2)

③ int b = random.nextInt(3)

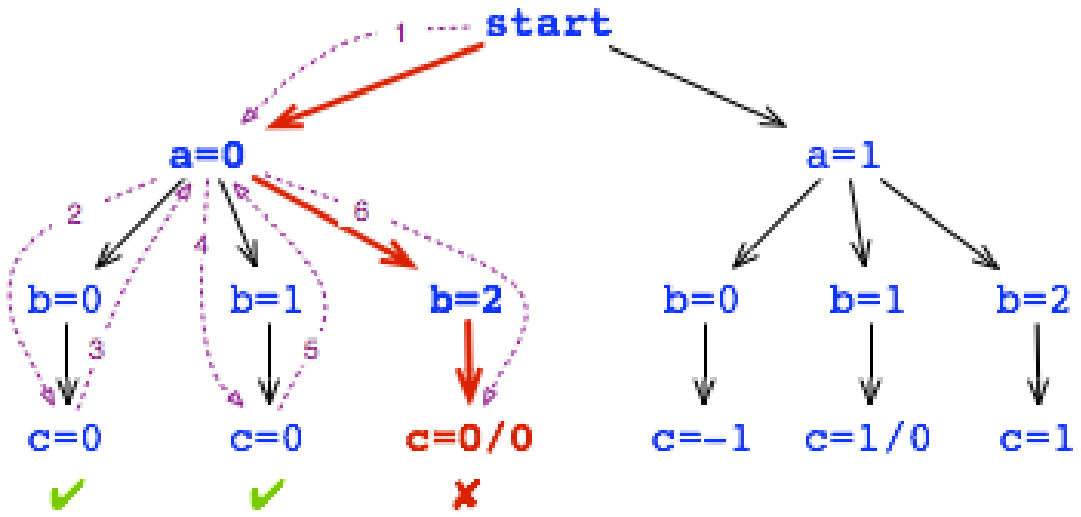
④ int c = a/(b+a -2)

What to do? Deal with it!

Given this code:

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```

Exhaustively search through all possible paths to find the defect!



① Random random = new Random()

② int a = random.nextInt(2)

③ int b = random.nextInt(3)

④ int c = a/(b+a -2)

Java Path Finder (JPF)

- Model checker developed by NASA to verify mission critical code

- Exhaustively searches through all possible states of a program
  - Enumerates all possible values from random number generators
  - Enumerates all possible interleavings between threads

JPF on Random Number Generation

```
int a = random.nextInt(2);
int b = random.nextInt(3);
int c = a/(b+a -2);
```

Not shown, but also generates a "trace" of random values chosen

```
-bash-4.2$ ./runJPF.sh Random.jpf
JavaPathfinder core system v8.0 (C) 2005-2014 United States Government.
…
====================================================== error 1
gov.nasa.jpf.vm.NoUncaughtExceptionsProperty
java.lang.ArithmeticException: division by zero
        at Rand.main(Rand.java:34)
…
```

Where in code exception happened

JPF on Thread Interleaving

```
-bash-4.2$ ./runJPF.sh Interleaving.jpf
JavaPathfinder core system v8.0 (C) 2005-2014 United States
Government.
…
====================================================== search started
main main Thread-1 Thread-1
main Thread-1 main Thread-1
main Thread-1 Thread-1 main
Thread-1 main main Thread-1
Thread-1 main Thread-1 main
Thread-1 Thread-1 main main
```

Able to explore all interleavings and generate all possible outputs!

Summary

Summary

- We learned there are two types of nondeterminism
  - Nondeterminism by mistake – stamp it out!
    - Memory errors
    - Datarace errors
  - Nondeterminism by design – deal with it!
    - Random number generation
    - Thread interleaving
- We also learned three tools that can help you
  - Google Address Sanitizer
  - Google Thread Sanitizer
  - NASA Java Path Finder

Open Source Resources

- Google Address Sanitizer:
  https://github.com/google/sanitizers/wiki/AddressSanitizer

- Google Thread Sanitizer:
  https://github.com/google/sanitizers/wiki/ThreadSanitizerCppManual

- NASA Java Path Finder:
  https://github.com/javapathfinder/jpf-core/wiki
  https://github.com/javapathfinder/jpf-core/wiki/GSoC-2022-Project-Ideas

References

- Konstantin Serebryany et al. “AddressSanitizer: A Fast Address Sanity Checker”. USENIX, 2012: https://research.google/pubs/pub37752/
- Konstantin Serebryany et al. “ThreadSanitizer – data race detection in practice”. Workshop on Binary Instrumentation and Applications (WBIA), 2009: https://research.google/pubs/pub35604/
- Ranjit Jhala and Rupak Majumdar. “Software model checking”. ACM Computing Surveys, 2009: https://people.mpi-sws.org/~rupak/Papers/SoftwareModelChecking.pdf
- 11th Competition on Software Verification (SV-COMP), 2022: https://sv-comp.sosy-lab.org/2022/results/results-verified/

Questions?